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CONNECTOR FOR HARDLINE COAXIAL CABLE

Abstract

A coaxial cable connector includes a nut housing having a rearward cable receiving end and a forward end opposite said rearward end, a front nut assembly coupled to the forward end of the nut housing, and a conductive metal tubular insert shaft supported within the nut housing or the front nut assembly. The front nut assembly includes an entry body housing and a conductive terminal pin extending from a forward end of the front nut assembly, and the conductive metal tubular insert shaft has a rearward end portion. A nonconductive plastic tubular support sleeve has a forward end portion coupled with the rearward end portion of the conductive metal tubular insert shaft, a tubular gripping ferrule radially surrounds the metal insert shaft and the plastic support sleeve, and a tubular outer sleeve radially surrounds at least a portion of said gripping ferrule. The gripping ferrule and the tubular outer sleeve are configured to be moved relative to one another in an axial direction such that the gripping ferrule and the tubular outer sleeve are configured to engage one another, thereby causing the gripping ferrule to radially compress around the conductive metal tubular insert shaft and the nonconductive plastic tubular support sleeve.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. Nonprovisional application Ser. No. 17/857,933 filed Jul. 5, 2022, pending, which is a continuation of U.S. Nonprovisional application Ser. No. 17/144,126 filed Jan. 7, 2021, now U.S. Pat. No. 11,381,028, which claims the benefit of U.S. Provisional Application No. 62/958,123, filed Jan. 7, 2020, and U.S. Provisional Application No. 63/031,597, filed May 29, 2020, the disclosures of which are hereby incorporated by reference herein in their entireties.

BACKGROUND

[0002] The present invention relates generally to connectors for terminating coaxial cable. More particularly, the present invention relates to axially compressible connectors for hardline or semi-rigid coaxial cables.

[0003] Coaxial cables are commonly used in the cable television industry to carry cable TV signals to television sets in homes, businesses, and other locations. A hardline coaxial cable may be used to carry the signals in distribution systems exterior to these locations and a flexible coaxial cable is then often used to carry the signals within the interior of these locations. Hardline or semi-rigid coaxial cable is also used where a high degree of radio-frequency (RF) shielding is required.

[0004] The hardline cable includes a solid wire core or inner conductor, typically of copper or copper-clad aluminum, surrounded by a solid tubular outer conductor. The outer conductor is also usually made of copper or aluminum. Dielectric material or insulation separates the inner and outer conductors. The outer conductor is covered with a cable jacket or sheath of plastic to provide protection against corrosion and weathering.

[0005] Threaded cable connectors, as shown in U.S. Pat. Nos. 5,352,134 and 6,019,636, have been employed to provide more even compression of the connector. Such connectors typically utilize some form of clamping mechanism that radially compresses the outer conductor of the cable against a tubular insert shaft upon axial threaded movement of the connector components to retain the cable in the hardline connector. The clamping mechanism may include a conical sleeve surrounded by an outer sleeve which forces the conical sleeve to radially compress upon axial movement of the outer sleeve with respect to the conical sleeve. The length of the conical closure sleeve typically closes the full length of the mechanism with equal forces around the circumference of the insert shaft. The resulting forces closing down on the coaxial cable compress the cable around the outside of the insert shaft creating a formed bond on the outside surface.

[0006] The ability of a connector to make a solid ground connection to the outer sheath of hardline CATV cables has always been required to achieve long term performance with respect to RFI shielding effectiveness of the connector as well as facilitate proper signal transmission through the

connector with minimal loss or disruption of said signal. Connectors throughout the CATV industry have been made with all metal mandrel support sleeves and also have been made with all plastic mandrel support sleeves. While the all metal holds up very well strength wise over time and temperature, the all plastic versions are susceptible to creep and can weaken over time and temperature.

[0007] There are varying levels of difficulties with different types of cable sold in the industry. For example, cables known as P3 or TX or T10 are often on the simpler side of things when it comes to making a solid ground connection between the cable and the connector. This is mostly due to the fact that all dielectric foam is removed from the inside of the outer conductor during the cable preparation process, prior to installing the connector. This removal of the dielectric foam allows for easy ground connection between the inside diameter of the cable and the outside diameter of the mandrel, which is typically made of a conductive metal. In the case of cables known as QR or even MC2, the cable preparation process leaves a thin film of non-conductive dielectric material on the inside diameter of the cable outer conductor. This layer prevents a solid ground connection being made as described above and seems to lengthen the signal pathway that the RF energy needs to travel as it propagates through the connector having a metal mandrel support sleeve, as shown in broken lines in FIG. 4. This extended pathway leads to the signal becoming out of phase and can cause “ringing” or harmonic in the signal response. This poor ground connection also leads to weakened RFI shielding performance and can also show up as a suckout or notch in the insertion loss performance of the connector.

[0008] It may be desirable to provide a connector that overcomes one or more of the aforementioned disadvantages of hardline connectors having an all metal or all plastic support sleeve. That is, it may be desirable to provide a connector having a hybrid metal-plastic support sleeve.

SUMMARY

[0009] In accordance with various embodiments of the present disclosure, a coaxial cable connector includes a nut housing having a rearward cable receiving end and a forward end opposite said rearward end, a front nut assembly coupled to the forward end of the nut housing, and a conductive metal tubular insert shaft supported within the nut housing or the front nut assembly. The front nut assembly includes an entry body housing and a conductive terminal pin extending from a forward end of the front nut assembly, and the conductive metal tubular insert shaft has a rearward end portion. A nonconductive plastic tubular support sleeve has a forward end portion coupled with the rearward end portion of the conductive metal tubular insert shaft, a tubular gripping ferrule radially surrounds the metal insert shaft and the plastic support sleeve, and a tubular outer sleeve radially surrounds at least a portion of said gripping ferrule. The gripping ferrule and the tubular outer sleeve are configured to be moved relative to one another in an axial direction such that the gripping ferrule and the tubular outer sleeve are configured to engage one another, thereby causing the gripping ferrule to radially compress around the conductive metal tubular insert shaft and the nonconductive plastic tubular support sleeve.

[0010] In some aspects, the conductive metal tubular insert shaft includes an engagement structure configured to engage an engagement structure of the nonconductive plastic tubular support sleeve to couple the conductive metal tubular insert shaft with the nonconductive plastic tubular support sleeve.

[0011] According to various aspects, the coaxial cable connector further includes a back nut assembly configured to be coupled with the rearward end of the nut housing, and the back nut assembly including an end cap. In some aspects, a mid nut assembly includes the nut housing, the nonconductive plastic tubular support sleeve, and the tubular gripping ferrule. In various aspects, the mid nut assembly further includes the conductive metal tubular insert shaft and the tubular outer sleeve.

[0012] According to some aspects, a back nut assembly includes the nut housing, the

nonconductive plastic tubular support sleeve, and the tubular gripping ferrule.

[0013] In some aspects, the front nut assembly includes the nonconductive plastic tubular support sleeve and the conductive metal tubular insert shaft.

[0014] In accordance with various embodiments of the present disclosure, a coaxial cable connector includes a nut housing having a rearward cable receiving end and an opposite forward end, a front nut assembly coupled to the forward end of the nut housing, a conductive metal tubular insert shaft supported within the nut housing or the front nut assembly, a nonconductive plastic tubular support sleeve having a forward end portion coupled with a rearward end portion of the conductive metal tubular insert shaft, a tubular gripping ferrule radially surrounding the metal insert shaft and the plastic support sleeve, and a tubular outer sleeve radially surrounding at least a portion of said gripping ferrule. The gripping ferrule and the tubular outer sleeve are configured to be moved relative to one another in an axial direction such that the gripping ferrule and the tubular outer sleeve are configured to engage one another, thereby causing the gripping ferrule to radially compress around the conductive metal tubular insert shaft and the nonconductive plastic tubular support sleeve.

[0015] According to some aspects, the conductive metal tubular insert shaft includes an engagement structure configured to engage an engagement structure of the nonconductive plastic tubular support sleeve to couple the conductive metal tubular insert shaft with the nonconductive plastic tubular support sleeve.

[0016] In various aspects, a back nut assembly is configured to be coupled with the rearward end of the nut housing and includes an end cap. According to some aspects, a mid nut assembly includes the nut housing, the nonconductive plastic tubular support sleeve, and the tubular gripping ferrule. In some aspects, the mid nut assembly further includes the conductive metal tubular insert shaft and the tubular outer sleeve.

[0017] According to some aspects, a back nut assembly includes the nut housing, the nonconductive plastic tubular support sleeve, and the tubular gripping ferrule.

[0018] In some aspects, the front nut assembly includes the nonconductive plastic tubular support sleeve and the conductive metal tubular insert shaft.

[0019] In various aspects, the front nut assembly includes an entry body housing and a conductive terminal pin extending from a forward end of the front nut assembly.

[0020] In accordance with various embodiments of the present disclosure, a coaxial cable connector includes a nut assembly having a rearward cable receiving end and an opposite forward end, a hybrid inner sleeve comprising a conductive forward portion and a nonconductive rearward portion, a tubular gripping ferrule radially surrounding the metal insert shaft and the plastic support sleeve, and a tubular outer sleeve radially surrounding at least a portion of said gripping ferrule. The gripping ferrule and the tubular outer sleeve are configured to be moved relative to one another in an axial direction such that the gripping ferrule and the tubular outer sleeve are configured to engage one another, thereby causing the gripping ferrule to radially compress around the hybrid inner sleeve.

[0021] According to some aspects, the conductive forward portion of the inner sleeve is a conductive metal tubular insert shaft, the conductive metal tubular insert shaft having a rearward end portion, and the nonconductive rearward portion is a nonconductive plastic tubular support sleeve having a forward end portion coupled with the rearward end portion of the conductive metal tubular insert shaft.

[0022] In some aspects, the conductive forward portion includes an engagement structure configured to engage an engagement structure of the nonconductive rearward portion to couple the conductive forward portion with the nonconductive rearward portion.

[0023] According to various aspects, a back nut assembly is configured to be coupled with the rearward end of the nut housing and includes an end cap. In some aspects, a mid nut assembly includes the nut housing, the nonconductive plastic tubular support sleeve, and the tubular gripping

ferrule. In various aspects, the mid nut assembly further includes the conductive metal tubular insert shaft and the tubular outer sleeve.

[0024] In various aspects, a back nut assembly includes the nut housing, the nonconductive plastic tubular support sleeve, and the tubular gripping ferrule.

[0025] According to some aspects, the coaxial cable connector further includes a front nut assembly configured to be coupled with the nut housing, the front nut assembly including the nonconductive plastic tubular support sleeve and the conductive metal tubular insert shaft.

[0026] In some aspects, the coaxial cable connector further includes a front nut assembly configured to be coupled with the nut housing, and the front nut assembly includes an entry body housing and a conductive terminal pin extending from a forward end of the front nut assembly.

[0027] Various aspects of the hardline coaxial connector, as well as other embodiments, objects, features and advantages of this disclosure, will be apparent from the following detailed description of illustrative embodiments thereof, which is to be read in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] FIG. 1 is an exploded perspective view of a conventional hardline connector.

[0029] FIG. 2 is a side cross-sectional view of the connector of FIG. 1.

[0030] FIG. 3 is an enlarged side cross-sectional view of the connector of FIG. 1.

[0031] FIG. 4 is a further enlarged side cross-sectional view of the connector of FIG. 1.

[0032] FIG. 5 is a side cross-sectional view of another conventional hardline connector.

[0033] FIG. 6 is an exploded perspective view of an exemplary hardline connector in accordance with various aspects of the disclosure.

[0034] FIG. 7 is a side cross-sectional view of the connector of FIG. 6.

[0035] FIG. 8 is an enlarged side cross-sectional view of the connector of FIG. 6.

[0036] FIG. 9 is an exploded perspective view of another exemplary hardline connector in accordance with various aspects of the disclosure.

[0037] FIG. 10 is a side cross-sectional view of the connector of FIG. 9.

[0038] FIG. 11 is an exploded perspective view of yet another exemplary hardline connector in accordance with various aspects of the disclosure.

[0039] FIG. 12 is a side cross-sectional view of the connector of FIG. 11.

[0040] FIG. 13 is an exploded perspective view of another exemplary hardline connector in accordance with various aspects of the disclosure.

[0041] FIG. 14 is a side cross-sectional view of the connector of FIG. 13.

DETAILED DESCRIPTION OF EMBODIMENTS

[0042] Referring first to FIGS. 1-4, a conventional connector 10 is depicted. The connector 10 is for hardline or semi-rigid coaxial cables. The connector 10 includes a front nut assembly 12 and a back nut assembly 14 that are configured to be removably connected to one another while providing both an electrical and mechanical connection therebetween.

[0043] As also shown in FIGS. 3 and 4, a coaxial cable 100 is inserted into the rearward end of the back nut assembly 14 of the connector 10. Coaxial cables 100 generally include a solid center conductor 102 typically formed from a conductive metal, such as copper, copper clad aluminum, copper clad steel, or the like capable of conducting electrical signals therethrough. Surrounding the cable center conductor 102 is a cable dielectric 104, which insulates the cable center conductor to minimize signal loss. The cable dielectric 104 also maintains a spacing between the cable center conductor 102 and a cable outer conductor or shield 106. The cable dielectric 104 is often a plastic material, such as a polyethylene, a fluorinated plastic material, such as a polyethylene or a

polytetrafluoroethylene, a fiberglass braid, or the like. The cable shield or outer conductor **106** is typically made of metal, such as aluminum or copper, and is often extruded to form a hollow tubular structure with a solid wall having a smooth exterior surface. An insulative cable jacket (not shown) may surround the cable outer conductor **106** to further seal the coaxial cable **100**. The cable jacket is typically made of plastic, such as polyvinylchloride, polyethylene, polyurethane, or polytetrafluoroethylene.

[0044] The connector **10** includes a plurality of components generally having a coaxial configuration about an axis defined by the center conductor **102** of the coaxial cable **100**. The front nut assembly **12** includes an entry body housing **16** supporting a terminal assembly **18** therein. Specifically, the entry body housing **16** is formed with an axial bore configured to cooperatively contain the terminal assembly **18** and is made from an electrically conductive material such as aluminum, brass or the like. The entry body housing **16** is formed with a threaded portion **20** at its forward end and a rearward threaded portion **22** opposite the forward threaded portion. The forward threaded portion **20** is configured to cooperate with devices located in the field that receive the forward end of the pin assembly **18**. An O-ring **24** may be provided around the forward threaded portion **30** to improve the seal that is made with a device and a portion of the exterior perimeter of the entry body housing **16** may be provided with a hexagonal shape to accommodate the use of tools during installation.

[0045] The rearward threaded portion **22** of the front nut assembly **12** is configured to cooperate with the back nut assembly **14**. Specifically, the rearward threaded portion **22** includes a rim face **26** that cooperates with an insert shaft **32** of the back nut assembly **14**, as will be described in further detail below.

[0046] The back nut assembly **14** of the connector **10** includes a nut housing **28** having an axial bore and a compression subassembly **30** rotatably supported within the axial bore. The compression subassembly **30** generally includes the insert shaft **32**, a holder sleeve **34**, a cable gripping ferrule **36**, and an O-ring **42** arranged in a coaxial relationship about the central axis of the back nut housing **28**. The cable jacket O-ring **42** improves the seal between the nut housing **28** and the cable **100** upon assembly.

[0047] The back nut housing **28** is made from an electrically conductive material, such as aluminum, brass, or the like, and includes a forward internally threaded portion **44** that cooperates with the rearward threaded portion **22** of the entry body housing **16** so that the two connector portions may be threadedly coupled together. The exterior surface of the back nut housing **28** is preferably provided with a hexagonal shape to accommodate the use of tools to facilitate such threaded coupling.

[0048] At its rearward end, the back nut housing **28** is formed with an axial bore **46** dimensioned to receive the outside diameter of the cable **100** in snug fitting relationship. At its forward end, opposite the rearward end, the back nut housing **28** is formed with a forward axial bore **47** communicating with the rearward axial bore **46** and dimensioned to accommodate the outer diameter of the insert shaft **32**. The back nut housing **28** is also preferably formed with an internal annular shoulder **48** that prevents rearward movement of the holder sleeve **34**, and thus the gripping ferrule **36**, as the gripping ferrule is radially compressed, as will be discussed in further detail below.

[0049] The insert shaft **32** includes a tubular body **52** terminating at a forward flanged head portion **54**. The insert shaft **32** is made from metal. The outside diameter of the tubular body **52** of the insert shaft **32** is dimensioned to be fitted within the inner diameter of the outer conductor **106** of the coaxial cable **100**. Also, the inside diameter of the tubular body **52** is dimensioned to provide a passageway to receive the center conductor **102** of the cable **100** after the cable has been prepared for termination, wherein a length of the dielectric **104** has been removed from the forward end of the cable.

[0050] The holder sleeve **34** is preferably made from an electrically conductive material, such as

aluminum or brass, and includes a sleeve body **58** having an exterior surface configured to be received within the forward axial bore **47** of the back nut housing **28**. The sleeve body **58** terminates at a rearward edge **60**, which engages the annular shoulder **48** of the back nut housing **28**.

[0051] The cable gripping ferrule **36** is generally in the form of a split tube having an axial gap **66** extending the full length of the ferrule. The gap **66** permits the diameter of the ferrule **36** to be reduced more easily so that the ferrule can be uniformly, radially compressed around the insert shaft **32** upon rearward axial movement of the insert shaft **32**, as will be discussed in further detail below. The inner surface **68** of the gripping ferrule is preferably provided with structure to enhance gripping of the outer surface of the cable. Such structure may include internal threads, teeth or some other form of textured surface.

[0052] As mentioned above, the outer surface of the cable gripping ferrule **36** is provided with a circumferential ramped portion **62**, which engages a forward end **70** of the holder sleeve **34**, opposite the rearward edge **60**, upon rearward axial movement of the insert shaft **32** to radially compress the gripping ferrule **36**. The ramped portion **62** defines a conical segment of the cable gripping ferrule **36** that tapers radially inwardly in the rearward direction. A rearward portion of the gripping ferrule **36** is received in an axial bore of the holder sleeve **34**.

[0053] Operation and installation of the connector **10** will now be described. Initially, the end of the coaxial cable **100** that is to be inserted into the rearward end of the back nut housing **28** is prepared in a conventional manner. In particular, cable preparation entails removing about 0.75 inch (19.05 mm.) of cable dielectric **104**, outer cable conductor **106** and cable jacket to expose a portion of the center conductor **102** that will engage the pin-terminal assembly **18** of the front nut assembly **12**. In addition, about 1.25 inches (31.75 mm.) of the cable dielectric **104** is removed from within the outer cable conductor **106** to provide clearance for the installation of the insert shaft **32**, and about 0.5 inch (12.70 mm.) of cable jacket is removed to make an electrical connection with the inside surface **68** of the cable gripping ferrule **36**. After the cable end is prepared, it is inserted into the back nut housing **28** so that the portion of the center conductor **102** engages the pin-terminal assembly **18**.

[0054] The back nut housing **28** is next threadedly coupled and rotated with respect to the front nut housing **16** to translate the front nut and back nut assemblies **12**, **14** together along their central axes. As the front nut and back nut assemblies **12**, **14** are translated closer together, the rim face **26** of the front nut housing **16** engages a forward shoulder **64** of the insert shaft **32** to translate the insert shaft **32** towards the rear of the back nut housing **28**. The interlocking mating surfaces of the front nut and back nut assemblies **12**, **14** cooperate to limit the amount of rotation between the front nut housing **16** and the back nut housing **28**.

[0055] The rearward translation of the insert shaft **32** causes the outer ramp portion **62** of the gripping ferrule **36** to engage the forward end **70** of the holder sleeve **34**, resulting in a radial compression of the ferrule **36**. The radial compression of the ferrule **36** reduces the overall diameter of the ferrule **36** and reduces the axial gap **66** of the ferrule so that the inner threaded surface **68** of the ferrule **36** bites down on the exposed portion of the outer cable conductor **106** and presses the conductor against the insert shaft **32**.

[0056] Referring now to FIG. 5, another conventional hardline connector **200** is illustrated. The connector **200** is similar to the conventional connector **10** described above, except that the metal insert shaft **32** is replaced with a forward holder sleeve **238** constructed of metal and a plastic insert shaft **232**. The forward holder sleeve **238** includes a radially inward lip **239** that engages a rearward facing shoulder **233** of the insert shaft **232** to limit rearward axial movement of the insert shaft **232** relative to the back nut housing **28** during assembly of the front nut and back nut housings **16**, **28**.

[0057] Referring now to FIGS. 6-8, an exemplary hardline connector **300** in accordance with various aspects of the disclosure is illustrated. The connector **300** includes a front nut assembly **312** and a back nut assembly **314** that are configured to be removably connected to one another while

providing both an electrical and mechanical connection therebetween. As also shown in FIGS. 7 and 8, a coaxial cable **100** is inserted into the rearward end of the back nut assembly **314** of the connector **300**.

[0058] The connector **300** includes a plurality of components generally having a coaxial configuration about an axis defined by the center conductor **102** of the coaxial cable **100**. The front nut assembly **312** includes an entry body housing **316** supporting a terminal pin assembly **318** therein. Specifically, the entry body housing **316** is formed with an axial bore configured to cooperatively contain the terminal pin assembly **318** and is made from an electrically conductive material such as aluminum, brass or the like. The entry body housing **316** is formed with a threaded portion **320** at its forward end and a rearward threaded portion **322** opposite the forward threaded portion. The forward threaded portion **320** is configured to cooperate with devices located in the field that receive the forward end of the pin assembly **318**. An O-ring **324** may be provided around the forward threaded portion **320** to improve the seal that is made with a device and a portion of the exterior perimeter of the entry body housing **316** may be provided with a hexagonal shape to accommodate the use of tools during installation.

[0059] The rearward threaded portion **322** of the front nut assembly **312** is configured to cooperate with the back nut assembly **314**. Specifically, the rearward threaded portion **322** includes a rim face **326** that cooperates with a conductive insert shaft **332** of the back nut assembly **314**, as will be described in further detail below.

[0060] The back nut assembly **314** of the connector **300** includes a back nut housing **328** having an axial bore and a compression subassembly **330** rotatably supported within the axial bore. The compression subassembly **330** generally includes the conductive insert shaft **332**, a holder sleeve **334**, a nonconductive support sleeve **335**, a cable gripping ferrule **336**, and an O-ring **342** arranged in a coaxial relationship about the central axis of the back nut housing **328**. The cable jacket O-ring **342** improves the seal between the nut housing **328** and the cable **100** upon assembly.

[0061] The back nut housing **328** is made from an electrically conductive material, such as aluminum, brass, or the like, and includes a forward internally threaded portion **344** that cooperates with the rearward threaded portion **322** of the entry body housing **316** so that the two connector portions may be threadedly coupled together. The exterior surface of the back nut housing **328** is preferably provided with a hexagonal shape to accommodate the use of tools to facilitate such threaded coupling.

[0062] At its rearward end, the back nut housing **328** is formed with an axial bore **346** dimensioned to receive the outside diameter of the cable **100** in snug fitting relationship. At its forward end, opposite the rearward end, the back nut housing **328** is formed with a forward axial bore **347** communicating with the rearward axial bore **346** and dimensioned to accommodate the outer diameter of the insert shaft **332**. For example, the internal surface of the back nut housing **328** may include an annular lip **321** and an annular shoulder **323** that define an annular groove **325** having an axial dimension. The annular groove **325** receives an annular projection **327** extending radially outward from an outer surface of the insert shaft **332** and permits axial movement of the insert shaft **332** relative to the back nut housing **328** within the axial dimension of the annular groove **325**. The back nut housing **328** is also preferably formed with an internal annular shoulder **348** that prevents rearward movement of the holder sleeve **334**, and thus the gripping ferrule **336**, as the gripping ferrule is radially compressed, as will be discussed in further detail below.

[0063] The insert shaft **332** includes a tubular body **352** terminating at a forward flanged head portion **354**. The insert shaft **332** is made from metal. The outside diameter of the tubular body **352** of the insert shaft **332** is dimensioned to be fitted within the inner diameter of the outer conductor **106** of the coaxial cable **100**. Also, the inside diameter of the tubular body **352** is dimensioned to provide a passageway to receive the center conductor **102** of the cable **100** after the cable has been prepared for termination, wherein a length of the dielectric **104** has been removed from the forward end of the cable.

[0064] The support sleeve **335** is a tubular body made from plastic. The outside diameter of the tubular body of the support sleeve **335** is dimensioned to be fitted within the inner diameter of the outer conductor **106** of the coaxial cable **100**. Also, the inside diameter of the tubular body of the support sleeve **335** is dimensioned to provide a passageway to receive the center conductor **102** of the cable **100** after the cable has been prepared for termination, wherein a length of the dielectric **104** has been removed from the forward end of the cable. A forward region of the support sleeve **335** includes a retention structure **337** configured to receive a complementary retention structure **339** at a rearward region of the insert shaft **332**. For example, as illustrated, the retention structure **337** may be an annular groove, and the retention structure **339** may be an annular projection. The retention structures **337**, **339** cooperate to limit or prevent relative axial movement between the insert shaft **332** and the support sleeve **335**. The support sleeve **335** may also include a forward facing annular shoulder **341** that can engage a rearward edge **342** of the insert shaft **332**. The plastic support sleeve **335** may have a thicker radial wall than the metal insert shaft **332**. The metal insert shaft **332** has an axial length that extends into the gripping ferrule **336**, but does not extend to the rearward axial bore **346**. The plastic support sleeve **335** has an axial length that extends from the metal insert shaft within the gripping ferrule **336** to the rearward axial bore **346**.

[0065] The holder sleeve **334** is preferably made from an electrically conductive material, such as aluminum or brass, and includes a sleeve body **358** having an exterior surface configured to be received within the forward axial bore **347** of the back nut housing **328**. The sleeve body **358** terminates at a rearward edge **360**, which engages the annular shoulder **348** of the back nut housing **328**.

[0066] The cable gripping ferrule **336** is generally in the form of a split tube having an axial gap **366** extending the full length of the ferrule. The gap **366** permits the diameter of the ferrule **336** to be reduced more easily so that the ferrule can be uniformly, radially compressed around the insert shaft **332** and the support sleeve **335** upon rearward axial movement of the insert shaft **332**, as will be discussed in further detail below. The inner surface **368** of the gripping ferrule is preferably provided with structure to enhance gripping of the outer surface of the cable. Such structure may include internal threads, teeth or some other form of textured surface.

[0067] As mentioned above, the outer surface of the cable gripping ferrule **336** is provided with a circumferential ramped portion **362**, which engages a forward end **370** of the holder sleeve **334**, opposite the rearward edge **360**, upon rearward axial movement of the insert shaft **332** and the support shaft **335** to radially compress the gripping ferrule **336**. The ramped portion **362** defines a conical segment of the cable gripping ferrule **336** that tapers radially inwardly in the rearward direction. A rearward portion of the gripping ferrule **336** is received in an axial bore of the holder sleeve **334**.

[0068] Operation and installation of the connector **300** will now be described. Initially, the end of the coaxial cable **100** that is to be inserted into the rearward end of the back nut housing **328** is prepared in a conventional manner. In particular, cable preparation entails removing about 0.75 inch (19.05 mm.) of cable dielectric **104**, outer cable conductor **106** and cable jacket to expose a portion of the center conductor **102** that will engage the pin-terminal assembly **318** of the front nut assembly **312**. In addition, about 1.25 inches (31.75 mm.) of the cable dielectric **104** is removed from within the outer cable conductor **106** to provide clearance for the installation of the insert shaft **332** and the support sleeve **335**, and about 0.5 inch (12.70 mm.) of cable jacket is removed to make an electrical connection with the inside surface **368** of the cable gripping ferrule **336**. After the cable end is prepared, it is inserted into the back nut housing **328** so that the portion of the center conductor **102** engages the pin-terminal assembly **318**.

[0069] The back nut housing **328** is next threadedly coupled and rotated with respect to the front nut housing **316** to translate the front nut and back nut assemblies **312**, **314** together along their central axes. As the front nut and back nut assemblies **312**, **314** are translated closer together, the rim face **326** of the front nut housing **316** engages a forward shoulder **364** of the insert shaft **332** to

translate the insert shaft **332** and the support sleeve **335** towards the rear of the back nut housing **328**. The interlocking mating surfaces of the front nut and back nut assemblies **312**, **314** cooperate to limit the amount of rotation between the front nut housing **316** and the back nut housing **328**. [0070] The rearward translation of the insert shaft **332** and support sleeve **335** causes the outer ramp portion **362** of the gripping ferrule **336** to engage the forward end **370** of the holder sleeve **334**, resulting in a radial compression of the ferrule **336**. The radial compression of the ferrule **336** reduces the overall diameter of the ferrule **336** and reduces the axial gap **366** of the ferrule so that the inner threaded surface **368** of the ferrule **336** bites down on the exposed portion of the outer cable conductor **106** and presses the conductor against the insert shaft **332** and the support sleeve **335**.

[0071] Referring now to FIGS. **9** and **10**, another exemplary hardline connector **400** in accordance with various aspects of the disclosure is illustrated. The connector **400** includes a front nut assembly **412**, a mid nut assembly **413**, and a back nut assembly **414** that are configured to be removably connected to one another while providing both an electrical and mechanical connection therebetween. Although not illustrated, the connector **400** is configured such that a coaxial cable **100** can be inserted into the rearward end of the back nut assembly **414** of the connector **400**.

[0072] The connector **400** includes a plurality of components generally having a coaxial configuration about an axis defined by the center conductor **102** of the coaxial cable **100**. The front nut assembly **412** includes an entry body housing **416** supporting a terminal pin assembly **418** therein. Specifically, the entry body housing **416** is formed with an axial bore configured to cooperatively contain the terminal pin assembly **418** and is made from an electrically conductive material such as aluminum, brass or the like. The entry body housing **416** is formed with a threaded portion **420** at its forward end and a rearward threaded portion **422** opposite the forward threaded portion. The forward threaded portion **420** is configured to cooperate with devices located in the field that receive the forward end of the pin assembly **418**. An O-ring **424** may be provided around the forward threaded portion **420** to improve the seal that is made with a device and a portion of the exterior perimeter of the entry body housing **416** may be provided with a hexagonal shape to accommodate the use of tools during installation.

[0073] The rearward threaded portion **422** of the front nut assembly **412** is configured to cooperate with the mid nut assembly **413**. Specifically, the rearward threaded portion **422** includes a rim face **426** that engages an insert shaft **432** of the mid nut assembly **413**.

[0074] The mid nut assembly **413** of the connector **400** includes a nut housing **428** having an axial bore and a compression subassembly **430** rotatably supported within the axial bore. The compression subassembly **430** generally includes the insert shaft **432**, a holder sleeve **434**, a support sleeve **435**, and a cable gripping ferrule **436**.

[0075] The back nut assembly **414** of the connector **400** includes an end cap **429**, an insert sleeve **443**, a first O-ring **442**, and a second O-ring **445** arranged in a coaxial relationship about the central axis of the mid nut housing **428**. The first O-ring **442** improves the seal between the end cap **429** and the cable **100** upon assembly, and the second O-ring **445** improves the seal between the end cap **429** and the mid nut housing **428**.

[0076] The mid nut housing **428** is made from an electrically conductive material, such as aluminum, brass, or the like, and includes a forward internally threaded portion **444** that cooperates with the rearward threaded portion **422** of the entry body housing **416** so that the two connector portions may be threadedly coupled together. Similarly, the end cap **429** may be made from an electrically conductive material, such as aluminum, brass, or the like, and includes a forward internally threaded portion **431** that cooperates with a rearward threaded portion **433** of the mid nut housing **428** so that the two connector portions may be threadedly coupled together. The exterior surface of the mid nut housing **428** and/or the end cap **429** are preferably provided with a hexagonal shape to accommodate the use of tools to facilitate such threaded coupling.

[0077] The end cap **429** and the insert sleeve **443** are formed with an axial bore **446** dimensioned to

receive the outside diameter of the cable **100** in snug fitting relationship. At a forward end of the mid nut housing **428**, opposite the end cap **429**, the mid nut housing **428** is formed with a forward axial bore **447** communicating with the rearward axial bore **446** and dimensioned to accommodate the outer diameter of the insert shaft **432**. The end cap **429** is preferably formed with an internal annular shoulder **448** that prevents rearward movement of the holder sleeve **434**, and thus the gripping ferrule **436**, as the gripping ferrule is radially compressed, as will be discussed in further detail below.

[0078] The insert shaft **432** includes a tubular body **452** terminating at a forward flanged head portion **454**. The insert shaft **432** is made from metal. The outside diameter of the tubular body **452** of the insert shaft **432** is dimensioned to be fitted within the inner diameter of the outer conductor **106** of the coaxial cable **100**. Also, the inside diameter of the tubular body **452** is dimensioned to provide a passageway to receive the center conductor **102** of the cable **100** after the cable has been prepared for termination, wherein a length of the dielectric **104** has been removed from the forward end of the cable.

[0079] The support sleeve **435** is a tubular body made from plastic. The outside diameter of the tubular body of the support sleeve **435** is dimensioned to be fitted within the inner diameter of the outer conductor **106** of the coaxial cable **100**. Also, the inside diameter of the tubular body of the support sleeve **435** is dimensioned to provide a passageway to receive the center conductor **102** of the cable **100** after the cable has been prepared for termination, wherein a length of the dielectric **104** has been removed from the forward end of the cable. In some aspects, the inside diameter of the tubular body of the support sleeve **435** may taper from the rear end toward the forward end, as shown in FIG. **10**.

[0080] A forward region of the support sleeve **435** includes a retention structure **437** configured to receive a complementary retention structure **439** at a rearward region of the insert shaft **432**. For example, as illustrated, the retention structure **437** may be an annular groove, and the retention structure **439** may be an annular projection. The retention structures **437**, **439** cooperate to limit or prevent relative axial movement between the insert shaft **432** and the support sleeve **435**. The support sleeve **435** may also include a forward facing annular shoulder **441** that can engage a rearward edge **453** of the insert shaft **432**. The plastic support sleeve **435** may have a thicker radial wall than the metal insert shaft **432**. The metal insert shaft **432** has an axial length that extends into the gripping ferrule **436**, but does not extend to the rearward axial bore **446**. The plastic support sleeve **435** has an axial length that extends from the metal insert shaft **432** within the gripping ferrule **436** to the rearward axial bore **446**.

[0081] The holder sleeve **434** is preferably made from an electrically conductive material, such as aluminum or brass, and includes a sleeve body **458** having an exterior surface configured to be received within the forward axial bore **447** of the mid nut housing **428**. The sleeve body **458** terminates at a rearward edge **460**, which engages the annular shoulder **448** of the end cap **429** and a forward end of the insert sleeve **443**.

[0082] The cable gripping ferrule **436** is generally in the form of a split tube having an axial gap **466** extending the full length of the ferrule. The gap **466** permits the diameter of the ferrule **436** to be reduced more easily so that the ferrule can be uniformly, radially compressed around the insert shaft **432** and the support sleeve **435** upon rearward axial movement of the insert shaft **432**. The inner surface **468** of the gripping ferrule is preferably provided with structure to enhance gripping of the outer surface of the cable. Such structure may include internal threads, teeth or some other form of textured surface.

[0083] As mentioned above, the outer surface of the cable gripping ferrule **436** is provided with a circumferential ramped portion **462**, which engages a forward end **470** of the holder sleeve **434**, opposite the rearward edge **460**, upon forward axial movement of the holder sleeve **434** to radially compress the gripping ferrule **436**. The ramped portion **462** defines a conical segment of the cable gripping ferrule **436** that tapers radially inwardly in the rearward direction. A rearward portion of

the gripping ferrule **436** is received in an axial bore of the holder sleeve **434**.

[0084] Operation and installation of the connector **400** will now be described. Initially, the end of the coaxial cable **100** that is to be inserted through the back nut assembly **414** and into the rearward end of the mid nut housing **428** is prepared in a conventional manner. The mid nut housing **428** is threadedly coupled and rotated with respect to the front nut housing **416** and the end cap **429** is threadedly coupled and rotated with respect to the mid nut housing **428** to translate the front nut and mid nut assemblies **412**, **413** together along their central axes. As the front nut and mid nut assemblies **412**, **413** are translated closer together, the internal annular shoulder **448** engages the holder sleeve **434** to translate the holder sleeve **434** in a forward axial direction relative to the gripping ferrule **436**. The interlocking mating surfaces of the front nut, mid nut, and back nut assemblies **412**, **413**, **414** cooperate to limit the amount of rotation between the front nut housing **416**, the mid nut housing **428**, and the end cap **429**.

[0085] The forward translation of the holder sleeve **424** causes the forward end **470** of the holder sleeve **434** to engage the outer ramp portion **462** of the gripping ferrule **436**, resulting in a radial compression of the ferrule **436**. The radial compression of the ferrule **436** reduces the overall diameter of the ferrule **436** and reduces the axial gap **466** of the ferrule so that the inner threaded surface **468** of the ferrule **436** bites down on the exposed portion of the outer cable conductor **106** and presses the conductor against the insert shaft **432** and the support sleeve **435**.

[0086] Referring now to FIGS. **11** and **12**, an exemplary hardline connector **500** in accordance with various aspects of the disclosure is illustrated. The connector **500** includes a front nut assembly **512** and a back nut assembly **514** that are configured to be removably connected to one another while providing both an electrical and mechanical connection therebetween. Although not illustrated, the connector **500** is configured such that a coaxial cable **100** can be inserted into the rearward end of the back nut assembly **514** of the connector **500**.

[0087] The connector **500** includes a plurality of components generally having a coaxial configuration about an axis defined by the center conductor **102** of the coaxial cable **100**. The front nut assembly **512** includes an entry body housing **516** supporting a terminal pin assembly **518** therein. Specifically, the entry body housing **516** is formed with an axial bore configured to cooperatively contain the terminal pin assembly **518** and is made from an electrically conductive material such as aluminum, brass or the like. The entry body housing **516** is formed with a threaded portion **520** at its forward end and a rearward threaded portion **522** opposite the forward threaded portion **520**. The forward threaded portion **520** is configured to cooperate with devices located in the field that receive the forward end of the pin assembly **518**. An O-ring **524** may be provided around the forward threaded portion **520** to improve the seal that is made with a device and a portion of the exterior perimeter of the entry body housing **516** may be provided with a hexagonal shape to accommodate the use of tools during installation.

[0088] The rearward threaded portion **522** of the front nut assembly **512** is configured to cooperate with the back nut assembly **514**. Specifically, the rearward threaded portion **522** includes a rim face **526** and ramped surface **527** that cooperates with a ramped surface of the gripping ferrule **536**, as will be described in further detail below.

[0089] The back nut assembly **514** of the connector **500** includes a nut housing **528** having an axial bore and a compression subassembly **530** rotatably supported within the axial bore. The compression subassembly **530** generally includes a holder sleeve **534**, a cable gripping ferrule **536**, and an O-ring **542** arranged in a coaxial relationship about the central axis of the back nut housing **528**. The cable jacket O-ring **542** improves the seal between the nut housing **528** and the cable **100** upon assembly.

[0090] The back nut housing **528** is made from an electrically conductive material, such as aluminum, brass, or the like, and includes a forward internally threaded portion **544** that cooperates with the rearward threaded portion **522** of the entry body housing **516** so that the two connector portions may be threadedly coupled together. The exterior surface of the back nut housing **528** is

preferably provided with a hexagonal shape to accommodate the use of tools to facilitate such threaded coupling.

[0091] At its rearward end, the back nut housing **528** is formed with an axial bore **546** dimensioned to receive the outside diameter of the cable **100** in snug fitting relationship. The back nut housing **528** is also preferably formed with an internal annular shoulder **548** that prevents rearward movement of the holder sleeve **534**, and thus the gripping ferrule **536**, as the gripping ferrule is radially compressed.

[0092] For example, the internal surface of the entry body housing **516** may include an annular lip **521** and an annular shoulder **533** that define an annular groove **525** having an axial dimension. The annular groove **525** receives an annular projection **527** extending radially outward from an outer surface of the insert shaft **532** and permits axial movement of the insert shaft **532** relative to the entry body housing **516** within the axial dimension of the annular groove **525**.

[0093] The insert shaft **532** includes a tubular body **552** terminating at a forward flanged head portion **554**. The insert shaft **532** is made from metal. The outside diameter of the tubular body **552** of the insert shaft **532** is dimensioned to be fitted within the inner diameter of the outer conductor **106** of the coaxial cable **100**. Also, the inside diameter of the tubular body **552** is dimensioned to provide a passageway to receive the center conductor **102** of the cable **100** after the cable has been prepared for termination, wherein a length of the dielectric **104** has been removed from the forward end of the cable.

[0094] The support sleeve **535** is a tubular body made from plastic. The outside diameter of the tubular body of the support sleeve **535** is dimensioned to be fitted within the inner diameter of the outer conductor **106** of the coaxial cable **100**. Also, the inside diameter of the tubular body of the support sleeve **535** is dimensioned to provide a passageway to receive the center conductor **102** of the cable **100** after the cable has been prepared for termination, wherein a length of the dielectric **104** has been removed from the forward end of the cable. In some aspects, the inside diameter of the tubular body of the support sleeve **535** may taper from the rear end toward the forward end, as shown in FIG. **12**.

[0095] A forward region of the support sleeve **535** includes a retention structure **537** configured to receive a complementary retention structure **539** at a rearward region of the insert shaft **532**. For example, as illustrated, the retention structure **537** may be an annular groove, and the retention structure **539** may be an annular projection. The retention structures **537**, **539** cooperate to limit or prevent relative axial movement between the insert shaft **532** and the support sleeve **535**. The support sleeve **535** may also include a forward facing annular shoulder **541** that can engage a rearward edge **553** of the insert shaft **532**. The plastic support sleeve **535** may have a thicker radial wall than the metal insert shaft **532**. The metal insert shaft **532** has an axial length that extends into the gripping ferrule **536**, but does not extend to the rearward axial bore **546**. The plastic support sleeve **535** has an axial length that extends from the metal insert shaft within the gripping ferrule **536** to the rearward axial bore **546**.

[0096] The holder sleeve **534** is preferably made from an electrically conductive material, such as aluminum or brass, and has an exterior surface configured to be received within the forward axial bore **547** of the back nut housing **528**. The holder sleeve **534** terminates at a rearward edge **560**, which engages the annular shoulder **548** of the back nut housing **528**.

[0097] The cable gripping ferrule **536** is generally in the form of a split tube having an axial gap **566** extending the full length of the ferrule. The gap **566** permits the diameter of the ferrule **536** to be reduced more easily so that the ferrule can be uniformly, radially compressed around the insert shaft **532** and the support sleeve **535** upon forward axial movement of the gripping ferrule **536**, as will be discussed in further detail below. The inner surface **568** of the gripping ferrule **536** is preferably provided with structure to enhance gripping of the outer surface of the cable. Such structure may include internal threads, teeth or some other form of textured surface.

[0098] As mentioned above, the outer surface of the cable gripping ferrule **536** is provided with a

circumferential ramped portion, which engages a rearward end **526** of the entry body housing **516**, upon forward axial movement of the gripping ferrule **536** to radially compress the gripping ferrule **536**. The ramped portion defines a conical segment of the cable gripping ferrule **536** that tapers radially inwardly in the forward direction. A rearward portion of the gripping ferrule **536** is received in an axial bore of the holder sleeve **534**.

[0099] Operation and installation of the connector **500** will now be described. Initially, the end of the coaxial cable **100** that is to be inserted through the back nut housing **528** is prepared in a conventional manner. The back nut housing **528** is next threadedly coupled and rotated with respect to the front nut housing **516** to translate the front nut and back nut assemblies **512**, **514** together along their central axes. As the front nut and back nut assemblies **512**, **514** are translated closer together, the holder sleeve **534** engages the gripping ferrule **536** to translate the gripping ferrule **536** in an axial direction relative to the rim face **526** of the rearward threaded portion **522** (i.e., inner sleeve) of the entry body housing **516**. The interlocking mating surfaces of the front nut and back nut assemblies **512**, **514** cooperate to limit the amount of rotation between the front nut housing **516** and the back nut housing **528**.

[0100] The relative translation between the inner sleeve of the entry body housing **516** gripping ferrule **536** causes the outer ramp portion **561** of the gripping ferrule **536** to engage the rim face **526** of the rearward threaded portion **522** (i.e., inner sleeve) of the entry body housing **516**, resulting in a radial compression of the ferrule **536**. The radial compression of the ferrule **536** reduces the overall diameter of the ferrule **536** and reduces the axial gap **566** of the ferrule so that the inner threaded surface **568** of the ferrule **536** bites down on the exposed portion of the outer cable conductor **106** and presses the conductor against the tubular body **552** of the insert shaft **532** and the support sleeve **535**.

[0101] Referring now to FIGS. **13** and **14**, another exemplary hardline connector **600** in accordance with various aspects of the disclosure is illustrated. The connector **600** includes a front nut assembly **612**, a mid nut assembly **613**, and a back nut assembly **614** that are configured to be removably connected to one another while providing both an electrical and mechanical connection therebetween. Although not illustrated, the connector **600** is configured such that a coaxial cable **100** can be inserted into the rearward end of the back nut assembly **614** of the connector **600**.

[0102] The connector **600** includes a plurality of components generally having a coaxial configuration about an axis defined by the center conductor **102** of the coaxial cable **100**. The front nut assembly **612** includes an entry body housing **616** supporting a terminal pin assembly **618** therein. Specifically, the entry body housing **616** is formed with an axial bore configured to cooperatively contain the terminal pin assembly **618** and is made from an electrically conductive material such as aluminum, brass or the like. The entry body housing **616** is formed with a threaded portion **620** at its forward end and a rearward threaded portion **622** opposite the forward threaded portion. The forward threaded portion **620** is configured to cooperate with devices located in the field that receive the forward end of the pin assembly **618**. An O-ring **624** may be provided around the forward threaded portion **620** to improve the seal that is made with a device and a portion of the exterior perimeter of the entry body housing **616** may be provided with a hexagonal shape to accommodate the use of tools during installation.

[0103] The rearward threaded portion **622** of the front nut assembly **612** is configured to cooperate with the mid nut assembly **613**. Specifically, the rearward threaded portion **622** includes a rim face **626** that cooperates with a nut housing **628** of the mid nut assembly **613**, as will be described in further detail below.

[0104] The mid nut assembly **613** of the connector **600** includes the nut housing **628** having an axial bore and a compression subassembly **630** rotatably supported within the axial bore. The compression subassembly **630** generally includes a holder sleeve **634**, a support sleeve **635**, and a cable gripping ferrule **636**.

[0105] The back nut assembly **614** of the connector **600** includes an end cap **629**, an insert sleeve

643, a first O-ring **642**, and a second O-ring **645** arranged in a coaxial relationship about the central axis of the mid nut housing **628**. The first O-ring **642** improves the seal between the end cap **629** and the cable **100** upon assembly, and the second O-ring **645** improves the seal between the end cap **629** and the mid nut housing **628**.

[0106] The mid nut housing **628** is made from an electrically conductive material, for example, a machined metal such as aluminum, brass, or the like, and includes a forward internally threaded portion **644** that cooperates with the rearward threaded portion **622** of the entry body housing **616** so that the two connector portions may be threadedly coupled together. Similarly, the end cap **629** may be made from an electrically conductive material, such as aluminum, brass, or the like, and includes a forward externally threaded portion **631** that cooperates with a rearward threaded portion **633** of the back nut housing **628** so that the two connector portions may be threadedly coupled together. The exterior surface of the back nut housing **628** and/or the end cap **629** are preferably provided with a hexagonal shape to accommodate the use of tools to facilitate such threaded coupling.

[0107] The end cap **629** and the insert sleeve **643** are formed with an axial bore **646** dimensioned to receive the outside diameter of the cable **100** in snug fitting relationship. At a forward end of the back nut housing **628**, opposite the end cap **629**, the back nut housing **628** is formed with a forward axial bore **647** communicating with the rearward axial bore **646**. The end cap **629** is preferably formed with an internal annular shoulder **648** that prevents rearward movement of the holder sleeve **634**, and thus the gripping ferrule **636**, as the gripping ferrule **636** is radially compressed, as will be discussed in further detail below.

[0108] The back nut housing **628** includes a tubular body **651** that forms the forward axial bore **647**, a forward flanged head portion **654** extending inward from the tubular body **651** of the back nut housing **628**, and a tubular portion **652** that extends axially from the forward flanged head portion **654** in a rearward direction. The tubular portion **652** is spaced radially inward from the tubular body **651**, and the outside diameter of the tubular portion **652** is dimensioned to be fitted within the inner diameter of the outer conductor **106** of the coaxial cable **100**. Also, the inside diameter of the tubular portion **652** is dimensioned to provide a passageway to receive the center conductor **102** of the cable **100** after the cable has been prepared for termination, wherein a length of the dielectric **104** has been removed from the forward end of the cable.

[0109] The support sleeve **635** is a tubular body made from plastic. The outside diameter of the tubular body of the support sleeve **635** is dimensioned to be fitted within the inner diameter of the outer conductor **106** of the coaxial cable **100**. Also, the inside diameter of the tubular body of the support sleeve **635** is dimensioned to provide a passageway to receive the center conductor **102** of the cable **100** after the cable has been prepared for termination, wherein a length of the dielectric **104** has been removed from the forward end of the cable. In some aspects, the inside diameter of the tubular body of the support sleeve **635** may taper from the rear end toward the forward end, as shown in FIG. 14.

[0110] A forward region of the support sleeve **635** includes a retention structure **637** configured to receive a complementary retention structure **639** at a rearward region of the tubular portion **652**. For example, as illustrated, the retention structure **637** may be an annular groove, and the retention structure **639** may be an annular projection. The retention structures **637**, **639** cooperate to limit or prevent relative axial movement between the tubular portion **652** and the support sleeve **635**. The support sleeve **635** may also include a forward facing annular shoulder **641** that can engage a rearward edge **653** of the tubular portion **652**. The plastic support sleeve **635** may have a thicker radial wall than the metal tubular portion **652**. The metal tubular portion **652** has an axial length that extends into the gripping ferrule **636** but does not extend to the rearward axial bore **646**. The plastic support sleeve **635** has an axial length that extends from the metal tubular portion **652** within the gripping ferrule **636** to the rearward axial bore **646**.

[0111] The holder sleeve **634** is preferably made from an electrically conductive material, such as

aluminum or brass, and includes a sleeve body **658** having an exterior surface configured to be received within the forward axial bore **647** of the back nut housing **628**. The sleeve body **658** includes a retention structure **674**, for example, an annular groove, at its outer surface, and the back nut assembly **628** includes a retention structure **676**, for example, an annular groove, at an inner surface of the tubular body **651**. The retention structures **674**, **676** are configured to receive a snap ring **672** such when the snap ring **672** is received in the retention structures **674**, **676**, the sleeve body **658** is axially fixed relative to the back nut assembly **628**. The sleeve body **658** terminates at a rearward edge **660**, which engages the annular shoulder **648** of the end cap **629** and a forward end of the insert sleeve **643**.

[0112] The cable gripping ferrule **636** is generally in the form of a split tube having an axial gap **666** extending the full length of the ferrule. The gap **666** permits the diameter of the ferrule **636** to be reduced more easily so that the ferrule can be uniformly, radially compressed around the tubular portion **652** and the support sleeve **635** upon forward axial movement of the support sleeve **635**. The inner surface **668** of the gripping ferrule **636** is preferably provided with structure to enhance gripping of the outer surface of the cable. Such structure may include internal threads, teeth, or some other form of textured surface.

[0113] As mentioned above, the outer surface of the cable gripping ferrule **636** is provided with a circumferential ramped portion **662**, which engages a forward end **670** of the holder sleeve **634**, opposite the rearward edge **660**, upon forward axial movement of the holder sleeve **634** to radially compress the gripping ferrule **636**. The ramped portion **662** defines a conical segment of the cable gripping ferrule **636** that tapers radially inwardly in the rearward direction. A rearward portion of the gripping ferrule **636** is received in an axial bore of the holder sleeve **634**.

[0114] Operation and installation of the connector **600** will now be described. Initially, the cable gripping ferrule **636**, the holder sleeve **634**, and the snap ring **672** are inserted into the rear end of the back nut housing **628** between the tubular body **651** and the tubular portion **652**, and a forward end of the support sleeve **635** is inserted into a rear end of the tubular portion **652**, as illustrated in FIG. 14.

[0115] An end of the coaxial cable **100** that is to be inserted into the rearward end of the back nut housing **628** is prepared in a conventional manner. In particular, cable preparation entails removing about 0.75 inch (19.05 mm.) of cable dielectric **104**, outer cable conductor **106** and cable jacket to expose a portion of the center conductor **102** that will engage the pin-terminal assembly **618** of the front nut assembly **612**. In addition, about 1.25 inches (31.75 mm.) of the cable dielectric **104** is removed from within the outer cable conductor **106** to provide clearance for the installation of the tubular portion **652** of the back nut housing **628**, and about 0.5 inch (12.70 mm.) of cable jacket is removed to make an electrical connection with the inside surface **668** of the cable gripping ferrule **636**. After the cable end is prepared, it is inserted through the back nut assembly **614** and into the back nut housing **628** so that the portion of the center conductor **102** engages the pin-terminal assembly **618**.

[0116] The end cap **629** is threadedly coupled and rotated with respect to the back nut housing **628** to translate the mid nut and back nut assemblies **613**, **614** together along their central axes. As the mid nut and back nut assemblies **613**, **614** are translated closer together, the end cap **629** and/or the insert sleeve **643** causes the forward end **670** of the holder sleeve **634** to engage the outer ramp portion **662** of the gripping ferrule **636**, resulting in a radial compression of the ferrule **636**. The radial compression of the ferrule **636** reduces the overall diameter of the ferrule **636** and reduces the axial gap **666** of the ferrule so that the inner threaded surface **668** of the ferrule **636** bites down on the exposed portion of the outer cable conductor **106** and presses the conductor against the tubular portion **652** of the back nut housing **628**.

[0117] The back nut housing **628** is threadably coupled and rotated with respect to the front nut housing **616** to translate the front nut and mid nut assemblies **612**, **613** together along their central axes. As the front nut and back nut assemblies **612**, **613** are translated closer together, the rim face

626 of the front nut housing **616** engages a forward surface **664** of the forward flanged head portion **654** of the back nut housing **628**. The mating surfaces of the front nut and mid nut assemblies **612**, **613** cooperate to limit the amount of rotation between the front nut housing **616** and the back nut housing **628**.

[0118] Although the illustrative embodiments of the present invention have been described herein with reference to the accompanying drawings, it is to be understood that the invention is not limited to those precise embodiments, and that various other changes and modifications may be effected therein by one skilled in the art without departing from the scope or spirit of the invention.

[0119] Various changes to the foregoing described and shown structures will now be evident to those skilled in the art. Accordingly, the particularly disclosed scope of the invention is set forth in the following claims.

Claims

1. A coaxial cable connector having a hybrid mandrel portion configured to enhance ground connection and radio frequency (RF) shielding of the connector comprising: a housing having a rearward cable receiving end and a forward end opposite said rearward end; a front portion configured to be coupled with the forward end portion of the housing portion; a conductive insert portion supported within the housing portion or the front portion, the conductive insert portion having a rearward end portion; a nonconductive tubular support portion having a forward end portion coupled with the rearward end portion of the conductive insert portion; a gripping portion configured to radially surround the insert portion and the support portion; an outer portion configured to radially surround at least a portion of said gripping portion; wherein the gripping portion and the outer portion are configured to be moved relative to one another in an axial direction such that the gripping portion and the outer portion are configured to engage one another, thereby causing the gripping portion to radially compress around the conductive insert portion and the nonconductive support portion; and wherein the nonconductive support portion and the conductive insert portion are configured to form a hybrid mandrel portion such that a signal pathway that radio frequency (RF) energy needs to travel along the conductive insert portion is shortened relative to a combined length of the hybrid mandrel portion so as to enhance ground connection and RF shielding of the connector.
2. The coaxial cable connector of claim 1, wherein the conductive insert portion includes an engagement structure configured to engage an engagement structure of the nonconductive support portion to couple the conductive insert portion with the nonconductive support portion.
3. The coaxial cable connector of claim 1, further comprising a back assembly portion configured to be coupled with the rearward end of the housing portion, wherein the back assembly portion includes an end cap.
4. The coaxial cable connector of claim 3, further comprising a mid assembly portion that includes the housing portion, the nonconductive support portion, and the gripping portion.
5. The coaxial cable connector of claim 4, wherein the mid assembly portion further includes the conductive insert portion and the outer portion.
6. The coaxial cable connector of claim 1, further comprising a back assembly portion that includes the housing portion, the nonconductive support portion, and the gripping portion.
7. The coaxial cable connector of claim 1, wherein the front portion includes the nonconductive support portion and the conductive insert portion.
8. A coaxial cable connector having a hybrid mandrel portion configured to enhance ground connection and radio frequency (RF) shielding of the connector comprising: a housing portion having a rearward cable receiving end portion and an opposite forward end portion; a front portion coupled to the forward end of the housing portion; a conductive metal portion supported within the housing portion or the front portion; a nonconductive support portion having a forward end portion

coupled with a rearward end portion of the conductive tubular shaft portion; a gripping portion surrounding the conducting insert portion and the nonconductive support portion and configured to be radially compressed around the conductive insert portion and the nonconductive support portion; and wherein the nonconductive support portion and the conductive insert portion are configured to form a hybrid mandrel portion such that a signal pathway that radio frequency (RF) energy needs to travel along the conductive shaft portion is shortened relative to a combined length of the hybrid mandrel portion so as enhance ground connection and RF shielding of the connector.

9. The coaxial cable connector of claim 8, wherein the conductive insert portion includes an engagement structure configured to engage an engagement structure of the nonconductive tubular support portion to couple the conductive insert portion with the nonconductive tubular support portion.

10. The coaxial cable connector of claim 8, further comprising a back nut assembly configured to be coupled with the rearward end of the housing portion, wherein the back nut assembly includes an end cap.

11. The coaxial cable connector of claim 10, further comprising a mid nut assembly that includes the housing portion, the nonconductive support portion, and the tubular gripping portion.

12. The coaxial cable connector of claim 11, wherein the mid nut assembly further includes the conductive insert portion and the outer portion.

13. The coaxial cable connector of claim 8, further comprising a back nut assembly that includes the housing portion, the nonconductive support portion, and the gripping portion.

14. The coaxial cable connector of claim 8, wherein the front nut assembly includes the nonconductive support portion and the conductive insert portion.

15. The coaxial cable connector of claim 8, wherein the front nut assembly includes an entry body housing and a conductive terminal pin extending from a forward end of the front portion.

16. A coaxial cable connector having a hybrid mandrel portion configured to enhance ground connection and radio frequency (RF) shielding of the connector comprising: a hybrid mandrel portion comprising a conductive forward portion and a nonconductive rearward portion, wherein the hybrid mandrel portion is configured to be supported within a housing portion; and wherein the nonconductive rearward portion and the conductive forward portion are configured such that a signal pathway that radio frequency (RF) energy needs to travel along the conductive forward portion is shortened relative to a combined length of the hybrid mandrel portion so as enhance ground connection and RF shielding of the connector.

17. The coaxial cable connector of claim 16, wherein the conductive forward portion comprises a conductive metal tubular insert shaft having a rearward end portion, and wherein the nonconductive rearward portion comprises a nonconductive plastic tubular support sleeve having a forward end portion configured to be coupled with the rearward end portion of the conductive metal tubular insert shaft.

18. The coaxial cable connector of claim 16, wherein the conductive forward portion includes an engagement structure configured to engage an engagement structure of the nonconductive rearward portion to couple the conductive forward portion with the nonconductive rearward portion.

19. The coaxial cable connector of claim 16, further comprising a back nut assembly configured to be coupled with the rearward end of the housing portion, wherein the back nut assembly includes an end cap.

20. The coaxial cable connector of claim 19, further comprising a mid nut assembly that includes the nut housing, the nonconductive support portion, and the tubular gripping portion.
